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Open Review in computer science

Elsevier grand challenge on executable papers

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Abstract

We present *Open Review*, a web-based platform aimed at stimulating executable papers by means of post-publication peer-review. Its goal is to bring computer science researchers to collaboratively build their work upon previous research results, in such a way that transparency, reproducibility and sustainability of research results are greatly improved. The main design goals of the platform are clarity, conciseness, and reproducibility. Its main features are to: (i) provide incentives for making research communities to participate, (ii) make papers executable by means of *boards*' annotations, without necessarily involving the authors of an article, and (iii) give snapshots of the current research state on any given article.

Keywords: Executable papers, reproducible research, peer-review, collaborative research, Elsevier Grand Challenge.

1. Towards a post-publication model for executable papers

The design of executable papers is becoming more and more necessary in computer science. Most research articles published nowadays in computer science present experimental results which are difficult or impossible to reproduce. Such research articles are strongly detrimental to the whole research community, since they not only fail to provide reusable blocks of research, but also make the bibliographical tasks of researchers much harder.

The Elsevier Executable Paper Grand Challenge aims at enhancing “how scientific information is used and communicated in computer- and computational sciences”. In our opinion, the key challenge in making papers executable consists in *stimulating authors* to publish their data and code. Indeed, from the perspective of authors willing to have papers published, sharing data and code before publication creates higher workload and risks of rejection [1]. Therefore, this abstract carefully addresses the question of *incentives* for researchers to make their research reproducible.

Our proposal in the direction of making executable papers, called *Open Review*, is original in that it aims at providing a way to make *all* articles executables, whether already published or not. Moreover, it does not necessarily involve authors (although it stimulates them to also participate). Therefore, we do not address technical solutions which forces authors to comply with coding/data formats that can be embedded within an article prior to publication. The rationale behind the Open Review platform is to make executable papers by relying on the ongoing reviewing

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efforts which almost always happen after an article is published: those of all researchers trying to reproduce the results of published paper. This abstract summarizes what the platform proposes, why it should work, and how it brings answers for making executable papers a reality.

2. Open Review : A post-publication web-based platform for reproducible research

2.1. Platform overview

Open Review is a web-based platform, aimed at providing a concise view of the current research state on an article, together with reproducible results. For each published article, a web page will serve as a reference for following advances in the research related to the article. Each user will be given a *board* per article, where they can briefly sum up their position. These *individual* boards will be limited in size, *e.g.*, 1000 characters, to enforce *clarity and conciseness* of the researcher's main contributions. Supplementary material will be available on each board, allowing to bring detailed opinions, results, comparisons, and code. The main innovation with the boards will be to allow readers interested in a paper to get a *quick overview of the positions of the research community on a paper*, together with *executable code*.

Since researchers' code, findings and opinions evolve over time, users will be free to update their board and/or supplementary material whenever they wish. In order to follow these evolutions, a timeline will be provided on each article page. When first loaded, an article page will present the latest boards from all contributors. The timeline will allow to trace back, in an intuitive way, the contributions and opinions of each user over time. Internally, the timeline feature will be implemented using a versioning system.

2.2. Incentives for using the platform

Rather than a sharp change in the reviewing and research practices, the proposed Open Review platform will address the challenges raised by executable papers in a smooth and realistic way. In a first stage, we expect the research community to bring code and additional experiments to existing papers. The incentive for researchers will be to show their involvement in different research topics, and to find rewards in providing the research community with results which will strengthen or mitigate previously claimed results.

As a concrete scenario of the use of the Open Review platform, let us consider the following example. A researcher has developed an algorithm, which has just been accepted for publication. His experimental results show improvements upon two previously proposed approaches, published in articles A and B. In order to stimulate interest of other researchers for his work, he creates a board on A and B articles, and attaches the code allowing to reproduce his results (*e.g.*, a Matlab script, and a text file that describes how to use the code). Authors of A and B may then use their own boards to provide additional results, which show for example that their approach is still competitive using a slightly different experimental setup. The benefits are clear: researchers gain visibility, research is accelerated, and the code added to boards makes paper executable. Moreover, as a side-effect of these interactions, executable contributions will be more objective, in that they will highlight different facets of a published paper.

We also plan to promote identification and participation by using reputation metrics that will quantify the contributions of users on the platform (for example, *karma* points, successfully used on questions and answers web platforms). Regarding identification, we believe that contributors should be free to choose whether they are anonymous or identified. On most social web sites, users usually choose to identify themselves in order to promote their contributions. The freedom of anonymity will be left for those who think that their opinion should not be related to their profile. We however believe that the proportion of such individuals is low enough to allow anonymity on the review platform. These features, allowing researchers to provide insights on published papers and promote their work, will be the main incentives for researchers to use the Open Review platform.

3. Answers brought to the executable papers challenge

We discuss in this section the additional challenges identified by Elsevier in making paper executables. For the sake of clarity, we grouped these challenges along the four following categories, and summarize our ideas/answers for each of them:

(i) *Executability/ Validation: Make articles truly interactive, allowing readers to manipulate the result space by interactively tuning data and parameters.* Our firm belief is that the choice of software tools and data formats should be left to the users. The main incentive for making the code truly reproducible will be that contributors benefit from : increased visibility and credit for derivative work. As a side-effect, we expect that contributors will choose whenever possible open source software and data formats. Ideally, “the results should be easily reproduced by an independent researcher with at most 15 min of user effort, requiring only standard, freely available tools” [2]. Without requiring such a nice entry to reproducibility, any contributions on how to reproduce results should be welcome [3].

(ii) *Compatibility/ Size: Adapt to heterogeneous user operating systems, and deal with experiments involving large-scale computers and very large file sizes.* By being web-based, Open Review provides an OS-agnostic way to store results and discuss their content. The reproducibility of results will however require the user to be able to run the software and load the data formats provided by contributors. Our point is similar to the executability challenge discussed above: the users should be free to choose a data format and a specific implementation in order to leave the platform open to the largest possible range of research practices. Users will however be encouraged to use, whenever possible, open source and OS-independent software to make their result as reproducible as possible. Finally, the question of experimental results which involve high computing resources is difficult to answer, and we do not have convincing answers for this challenge. We simply mention that Open Review could help in giving links to platforms where data intensive experiments may be reproduced.

(iii) *Copyright/ Plagiarism: Maintain authors intellectual property.* By keeping track of each user’s contributions over time, the proposed open review system will allow to better identify *a posteriori* those who proposed results which help in advancing knowledge. We therefore believe that Open Review would help in bringing credit for any stakeholder in a research process, from the data generation to the resulting scientific outcomes. Regarding plagiarism, we believe that the platform will potentially serve as a place to detect such practices. More generally, intellectual property is an important asset in scientific research. If data or implementations must be kept confidential, authors will not be able to provide them on the platform (but they can still mention this on their boards). However, an interesting possibility will be for other users to provide alternative data sets and implementations which will still bring some executability to the paper. The license under which contributions are provided will be left to the user’s choice [4].

(iv) *Provenance/ Long-term compatibility: Track actions taken on executable papers, and make it compatible to future systems.* The tracking of boards over time is a planned feature of Open Review (see Section 2.1). The compatibility issue for the data formats and code will not be handled directly, but dealt with as a side-effect of the users’ care in having their contributions reproducible and sustainable. Regarding the evolution of the Open Review platform itself, we plan to make it open source, so that the code will be reusable and adaptable to different contexts.

4. Prototype development and perspectives

A prototype is currently being developed at the Vrije Universiteit of Brussels/Université Libre de Bruxelles (VUB/ULB), Belgium, which will be made open source during the first semester of 2011. The prototype will first be made available to a restrained research community of the VUB/ULB in order to thoroughly test the validity of the executable paper model presented here. The platform will then be made open to a wider range of universities, with possible enforcement of users’ identification in a first stage.

The proposed prototype answers the executable papers challenge in an original way, since the executability is not *embedded* in the paper, but made available by means of a *post-publication collaborative research platform*. We believe that our approach forms part of the what the next generation of executable papers will be. We stress that careful attention will be brought to make it interoperable with institutional repositories, publishing houses, and other digital libraries, using as a primary tool the Open Archives Initiative [5]. If successful, the main interest of the platform will be to bring executability to the widest possible range of articles, whether published or not.

5. Bibliography

- [1] J. Mesirov, Accessible reproducible research, *Science* 327 (5964) (2010) 415.
- [2] P. Vandewalle, J. Kovacevic, M. Vetterli, Reproducible research in signal processing, *Signal Processing Magazine, IEEE* 26 (3) (2009) 37–47.
- [3] N. Barnes, Publish your computer code: it is good enough, *Nature* 467 (7317) (2010) 753.
- [4] V. Stodden, The legal framework for reproducible scientific research: Licensing and copyright, *Computing in Science & Eng.* (2009) 35–40.
- [5] Open Archives Initiative, <http://www.openarchives.org>.